DEPARTMENT OF COMPUTER & INFORMATION SYSTEMS ENGINEERING
BACHELORS IN COMPUTER SYSTEMS ENGINEERING
Course Code: CS-324
Course Title: Machine Learning
Complex Engineering Problem
TE Batch 2022, Spring Semester 2025
Grading Rubric
TERM PROJECT

Group Members:

Student No.	Name	Roll No.
S1	Wajeeha Ali	CS-22055
S2	Tooba Fatima	CS-22056
S3	Asiya Fatima Tabassum	CS-22064

CRITERIA AND SCALES				Marks Obtained		
				S1	S2	S3
Criterion 1: Does the application meet the desired specifications and produce the desired outputs? (CPA-1, CPA-2, CPA-3) **[8 marks]**						
1	2	3	4			
The application does not meet the desired specifications and is producing incorrect outputs.	The application partially meets the desired specifications and is producing incorrect or partially correct outputs.	The application meets the desired specifications but is producing incorrect or partially correct outputs.	The application meets all the desired specifications and is producing correct outputs.			
Criterion 2: How well is the code organization? **[2 marks]**						
1	2	3	4			
The code is poorly organized and very difficult to read.	The code is readable only to someone who knows what it is supposed to be doing.	Some part of the code is well organized, while some part is difficult to follow.	The code is well organized and very easy to follow.			
Criterion 3: Does the report adhere to the given format and requirements? **[6 marks]**						
1	2	3	4			
The report does not contain the required information and is formatted poorly.	The report contains the required information only partially but is formatted well.	The report contains all the required information but is formatted poorly.	The report contains all the required information and completely adheres to the given format.			
Criterion 4: How does the student performed individually and as a team member? (CPA-1, CPA-2, CPA-3) **[4 marks]**						
1	2	3	4			
The student did not work on the assigned task.	The student worked on the assigned task, and accomplished goals partially.	The student worked on the assigned task, and accomplished goals satisfactorily.	The student worked on the assigned task, and accomplished goals beyond expectations.			

Final Score = (Criteria 1 score) + (Criteria 2 score) + (Criteria 3 score) + (Criteria 4 score)
= _____

_____
Teacher's Signature

INTRODUCTION:

The purpose of this research is to create a machine learning model that can predict weather conditions based on previous data. The dataset includes a variety of weather-related parameters such as temperature, humidity, wind speed, and visibility, as well as a summary of each observation. The primary goal is to organize the weather report into three simple categories: clear, cloudy, and precipitation. To do this, various machine learning models, such as K-Nearest Neighbors (KNN), Logistic Regression, and Artificial Neural Networks (ANN), were constructed and tested. The study focuses on data preparation, feature engineering, and model comparison to determine which machine learning algorithm performs best in predicting weather conditions using the supplied features.

DATA CLEANING AND PRE-PROCESSING:

In this phase, we clean the dataset by deleting extraneous columns, encoding categorical data, and normalizing the features. We also use SMOTE and outlier removal techniques to ensure that the data is appropriate for model training. These actions are critical to improve model performance and accuracy.

- *Data Loading and Exploration*

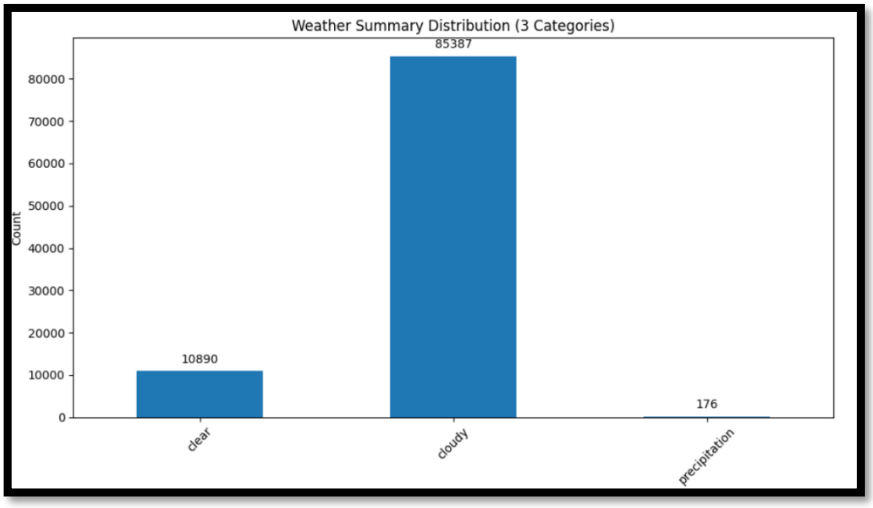
  The weather history dataset was put into pandas from a CSV file called weatherHistory.csv. The dataset was examined to better understand its structure and determine the necessary preprocessing measures.

- *Data Cleaning*

  The unnecessary columns such as Summary, Formatted Date, Loud Cover, Daily Summary, and Precip Type were dropped from the dataset, as they were deemed irrelevant to the predictive models.

- *Feature Engineering*

  A custom function *simplify_summary* was applied to the Summary column to map the original weather conditions to three simplified categories clear, cloudy, and precipitation. The simplified weather was encoded as clear = 0, cloudy = 1, precipitation = 2.

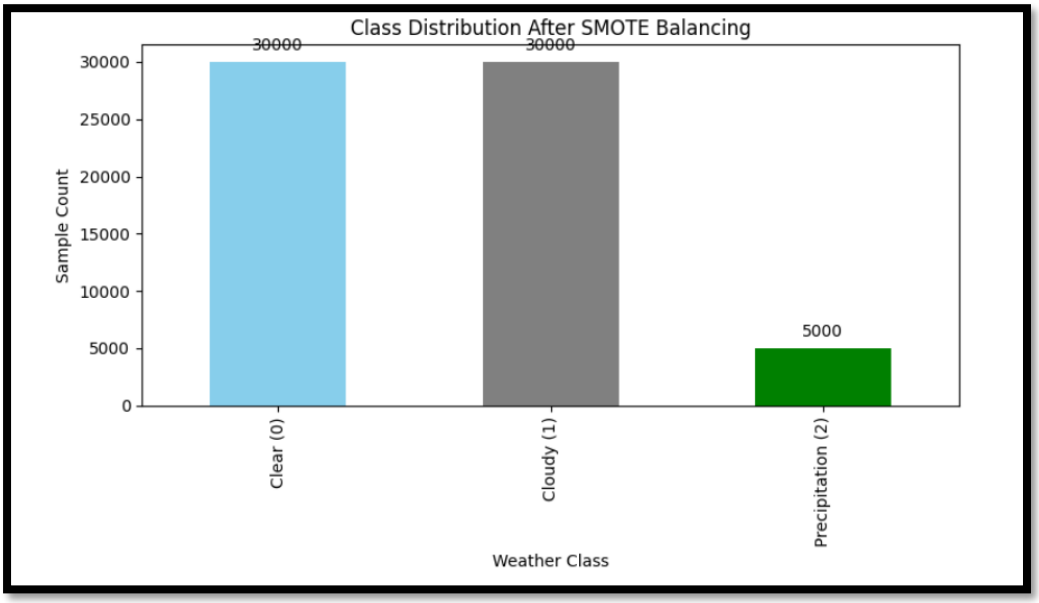


- *Data Scaling and Normalization*

  The feature columns were scaled using *StandardScaler* to standardize the data. This was essential for algorithms that rely on distance metrics, such as K-Nearest Neighbors (KNN).

- *Handling Imbalanced Data*

  The sample was unevenly distributed, with the cloudy class having the largest frequency. A down sampling technique was used to lower the number of samples in the cloudy class to 30,000, which improved the dataset's balance. The SMOTE was used on the dataset to balance the classes. The target distribution following SMOTE was set to Clear = 30,000 samples, cloudy represents 30,000 samples, precipitation = 5000 samples.



- *Outlier Removal*

  Outliers were detected using the Interquartile Range (IQR) method. Data points outside the range of 1.5 times the IQR were considered outliers and removed to ensure that the model wouldn't be affected by extreme values.

- *Data Splitting*

  The data was split into features (X) and target (y), with the target being the encoded weather summary class *Simple_Summary_Class*. The dataset was further split into training and test sets using a stratified split to maintain the distribution of classes in both subsets.

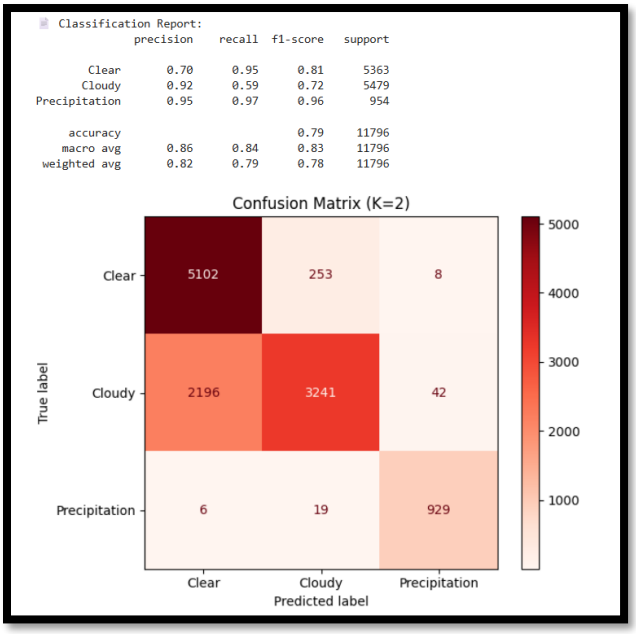
MODELS AND MACHINE LEARNING ALGORITHMS:

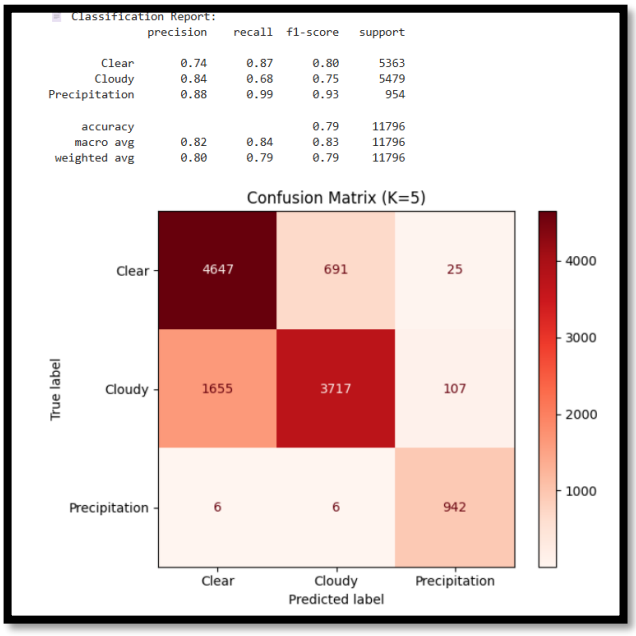
K-Nearest Neighbor (KNN):

The K-Nearest Neighbors (KNN) classifier is a non-parametric algorithm whose performance in forecasting weather conditions was evaluated using various k values. Three values of k were tested. The model was trained on scaled training data for each value of k, and its performance was evaluated through accuracy scores, confusion matrices, and classification reports. Evaluation metrics were recorded for both the training and testing sets to assess model performance, highlight potential overfitting or underfitting, and display the model's predictions using confusion matrices.

Model	Hyperparameters	Train Accuracy	Test Accuracy
1	k = 2	87.96%	78.60%
2	k = 5	85.76%	78.89%
3	k = 15	79.93%	77.02%

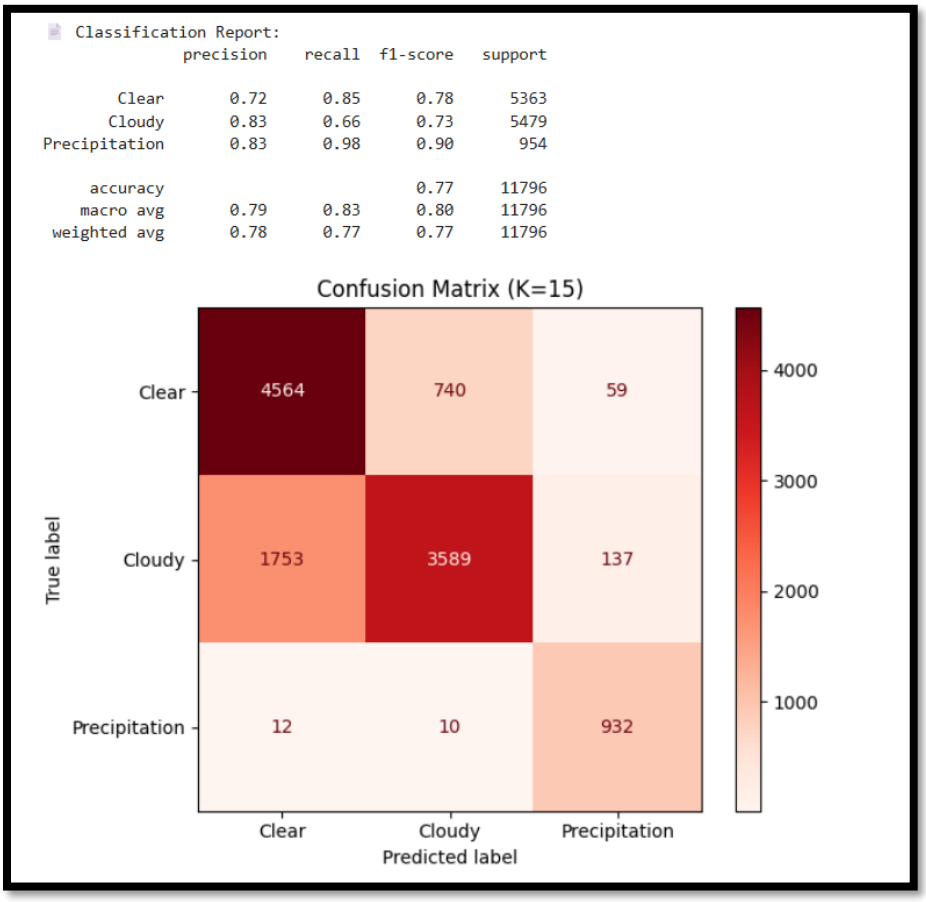
Confusion Matrix & classification report:



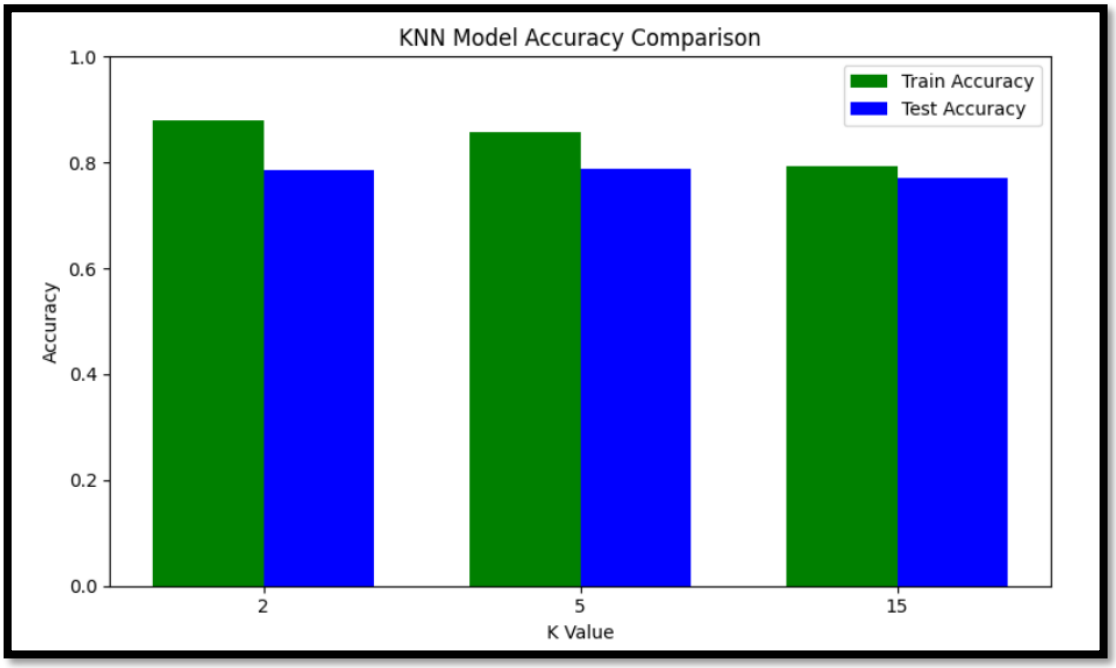


Classification Report:

	precision	recall	f1-score	support
Clear	0.72	0.85	0.78	5363
Cloudy	0.83	0.66	0.73	5479
Precipitation	0.83	0.98	0.90	954
accuracy			0.77	11796
macro avg	0.79	0.83	0.80	11796
weighted avg	0.78	0.77	0.77	11796



Graphical Representation:

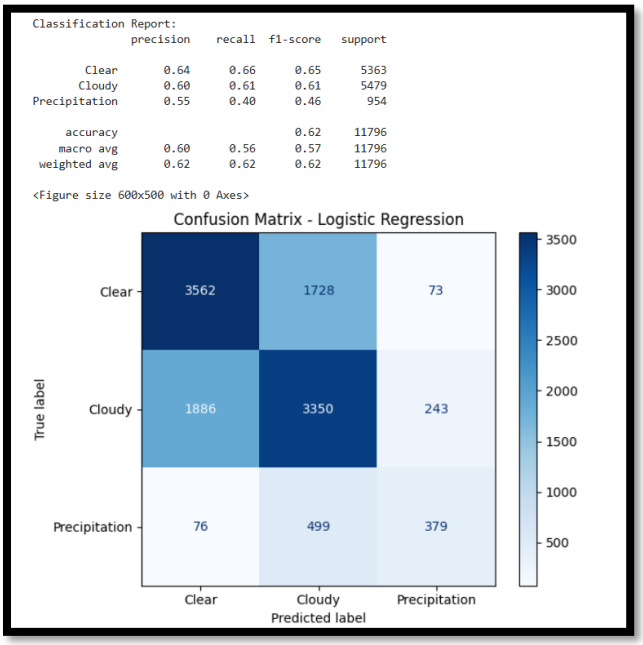


Logistic Regression:

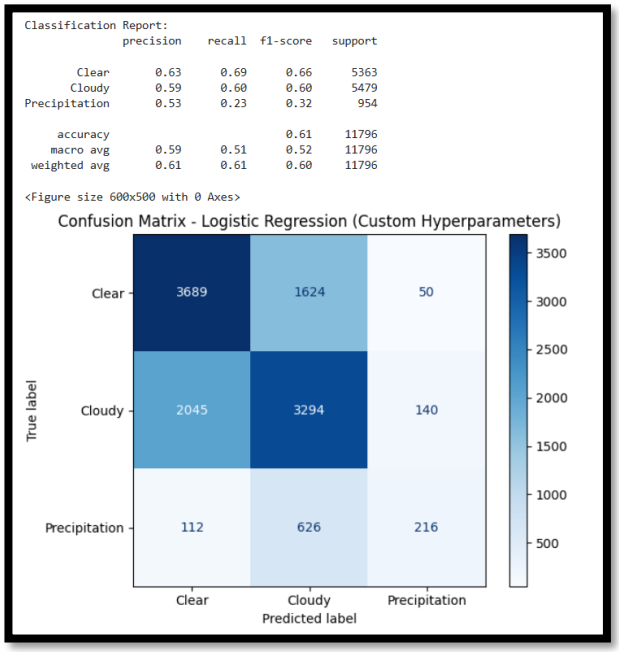
Three variations of the Logistic Regression model were implemented with different sets of hyperparameters to evaluate their performance on the weather classification task. The first model used the multinomial approach with the lbfgs solver and a maximum of 1000 iterations. The second model utilized the one-vs-rest (OVR) strategy with saga solver and L1 regularization to handle sparse features, with a regularization strength of 0.5 and 2000 iterations. The third model employed the multinomial strategy again but with the newton-cg solver and L2 regularization for improved multiclass probability modeling, along with a regularization strength of 1.5 and 3000 iterations to ensure convergence. For each model, predictions were made on the test data, and performance was evaluated using accuracy scores, classification reports, and confusion matrices. These metrics provided insights into the models' ability to correctly classify the weather conditions, with comparisons drawn between the different hyperparameter configurations.

Model	Hyperparameters	Train Accuracy	Test Accuracy
1	Multi-class = 'multinomial', solver = 'lbfgs', max_iter = 1000	61.20%	61.81%
2	Multi-class = 'ovr', solver = 'saga', penalty = 'l1', C = 0.5, max_iter = 2000	60.81%	61.03%
3	Multi-class = 'multinomial', solver = 'newton-cg', penalty = 'l2', C = 1.5, max_iter = 3000	61.21%	61.80%

Confusion Matrix & Classification Report:



Model 1



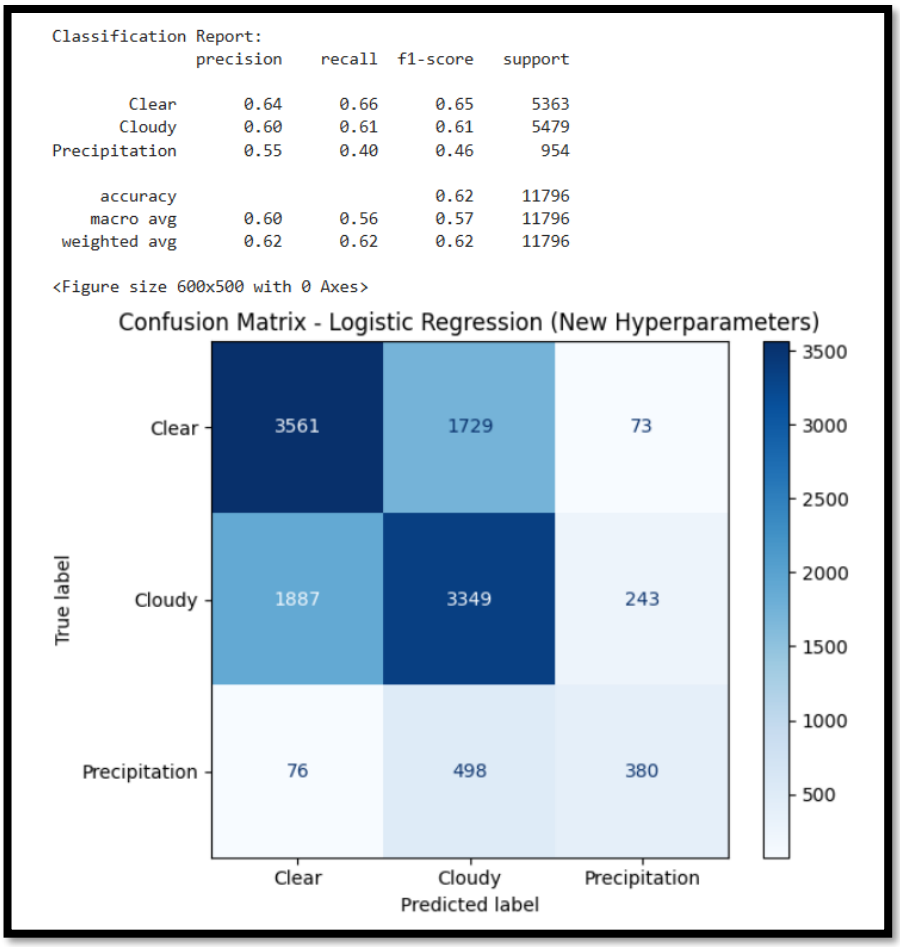
Model 2

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Classification Report:
               precision    recall  f1-score   support

        Clear       0.64      0.66      0.65      5363
       Cloudy       0.60      0.61      0.61      5479
Precipitation       0.55      0.40      0.46       954

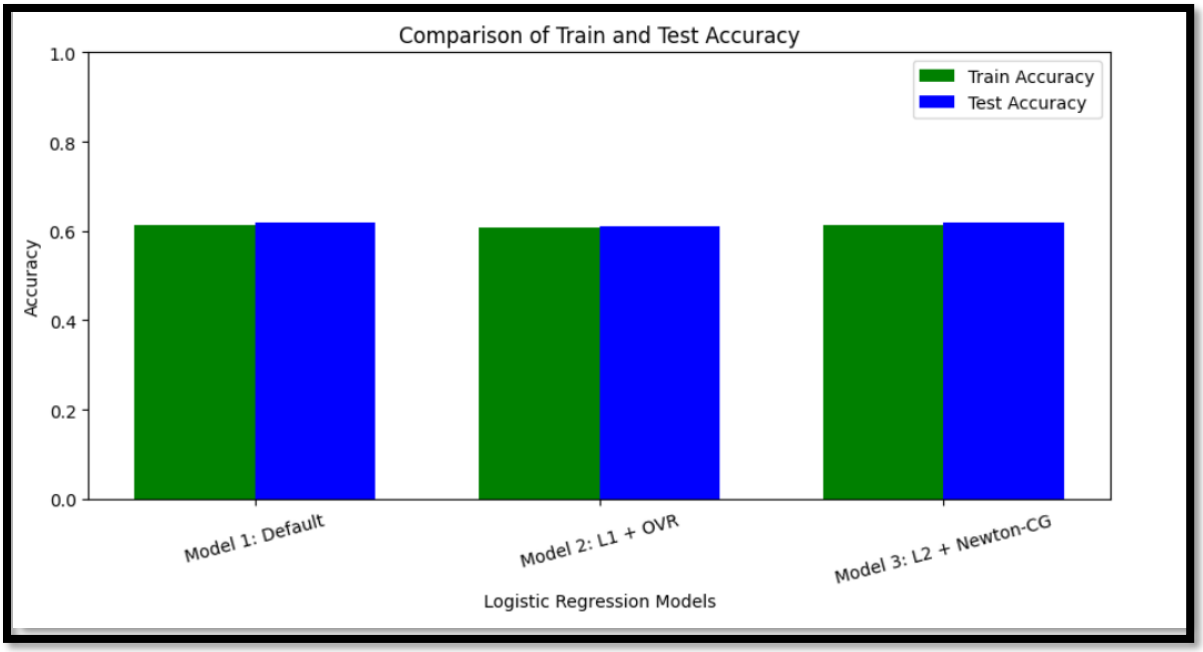
     accuracy                           0.62     11796
    macro avg       0.60      0.56      0.57     11796
 weighted avg       0.62      0.62      0.62     11796

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Model 3

Graphical Representation:

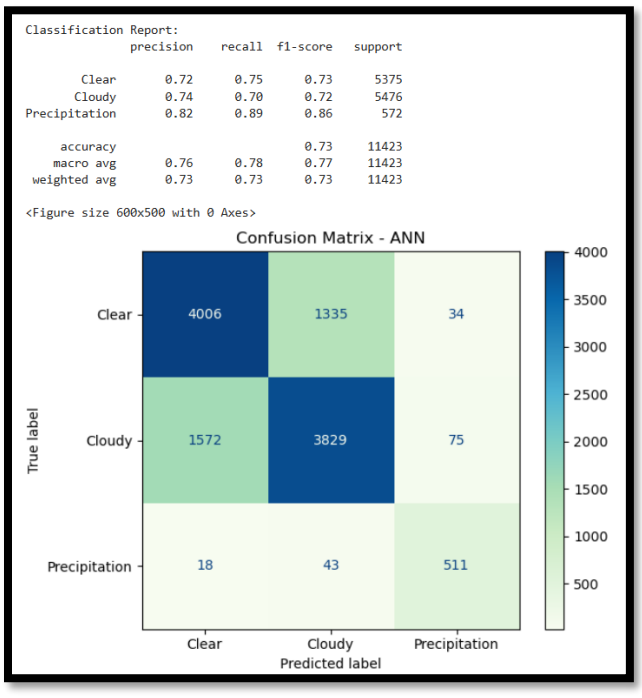


Artificial Neural Network (ANN):

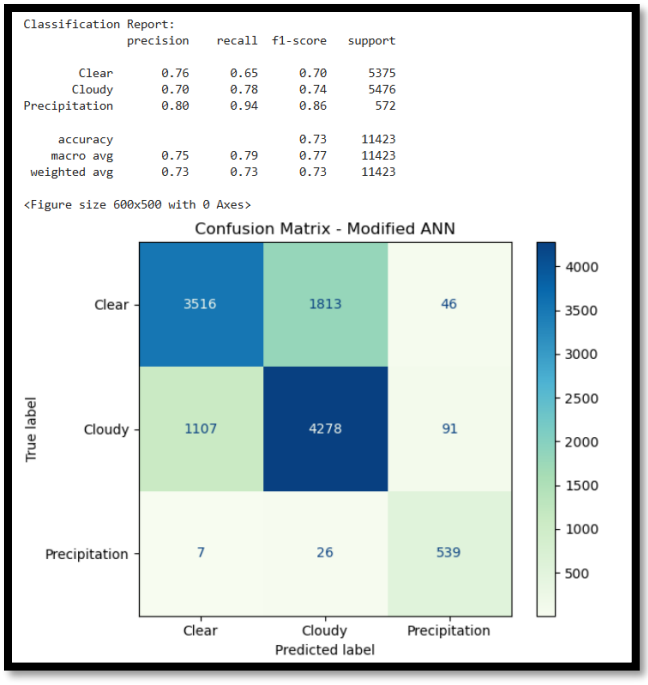
The Artificial Neural Network (ANN) was implemented with three configurations to predict weather conditions based on various features. The first model featured a simple architecture with 100 units in each of the two hidden layers, along with L2 regularization to help prevent overfitting. This model was trained for 20 epochs using a batch size of 100. After training, performance was evaluated using accuracy, loss, and confusion matrices for both the training and test datasets. In the second configuration, the architecture remained largely unchanged, but the training duration was increased to 50 epochs with a batch size of 130 and in third model extended training to 80 epochs with a batch size of 150. Throughout the process, classification reports and confusion matrices were generated to assess the model's ability to classify each weather condition correctly.

Model	Hyperparameters	Train Accuracy	Test Accuracy
1	Epochs = 20 & batch_size = 100	73.1%	72.4%
2	Epochs = 50 & batch_size = 130	73.9%	73.2%
3	Epochs = 80 & batch_size = 150	77.1%	75.2%

Confusion Matrix & Classification Report:



Model 1

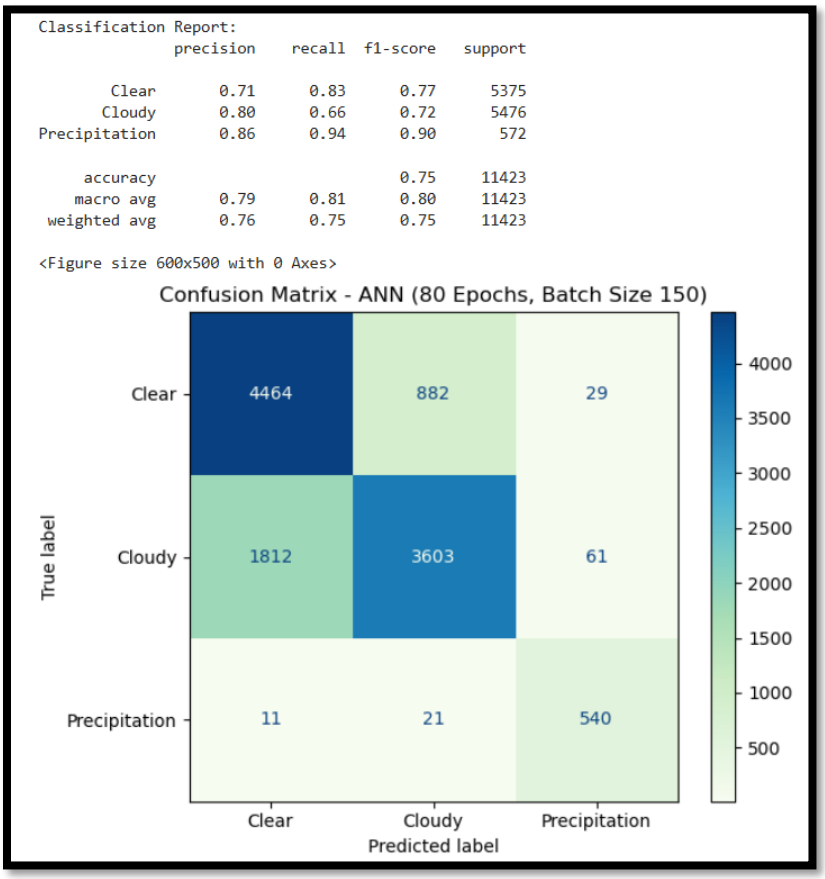


Model 2

Classification Report:

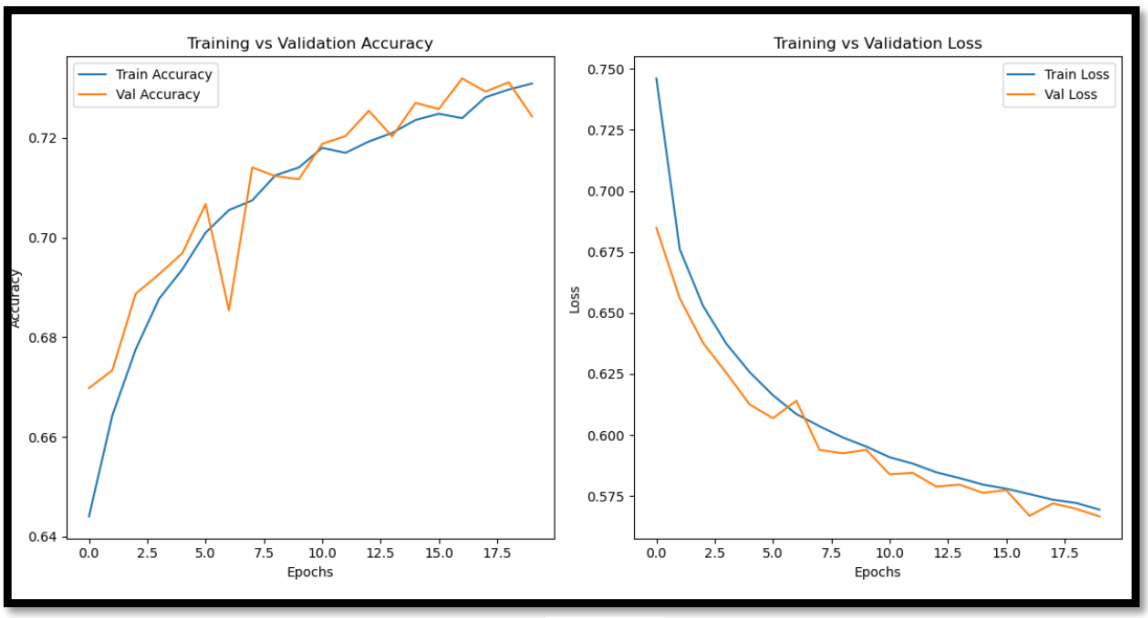
	precision	recall	f1-score	support
Clear	0.71	0.83	0.77	5375
Cloudy	0.80	0.66	0.72	5476
Precipitation	0.86	0.94	0.90	572
accuracy			0.75	11423
macro avg	0.79	0.81	0.80	11423
weighted avg	0.76	0.75	0.75	11423

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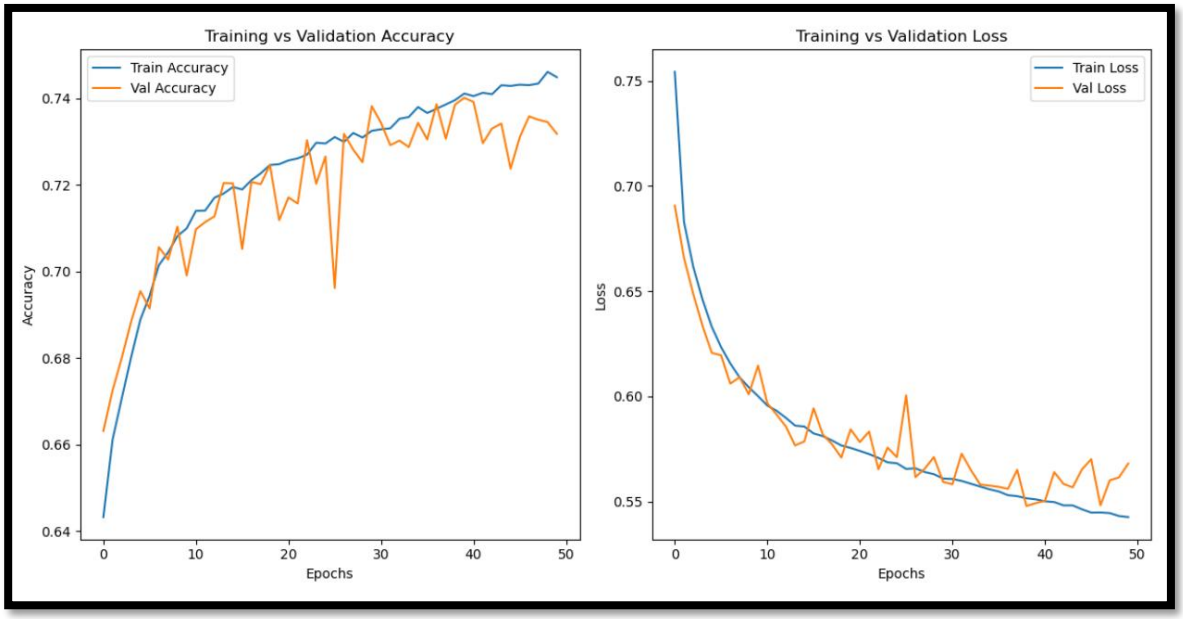


Model 3

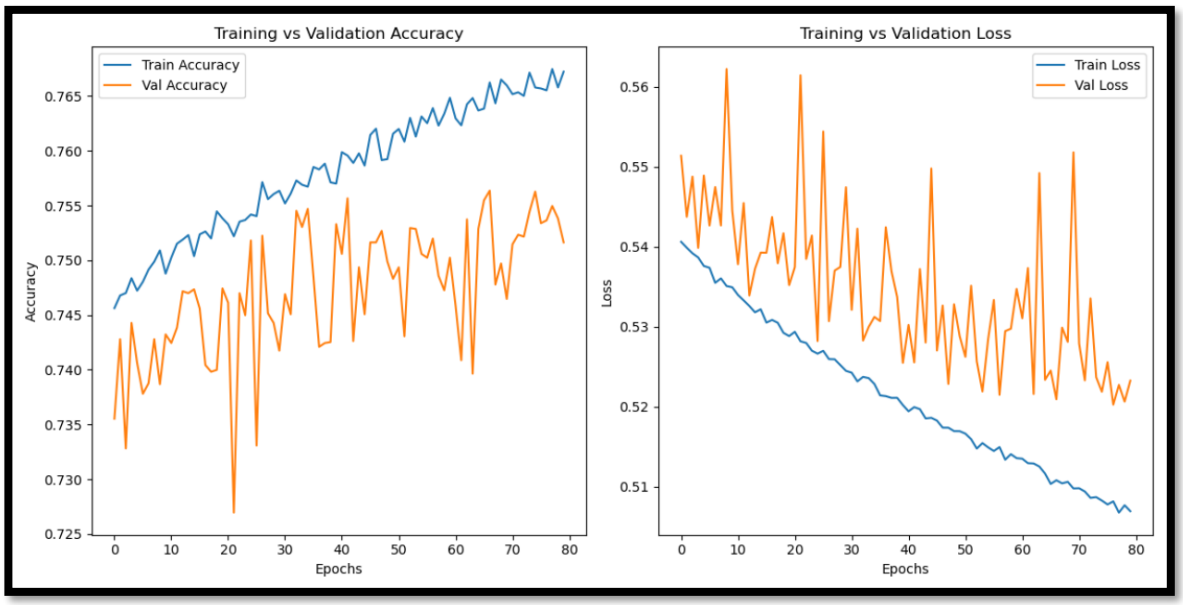
Graphical Representation:



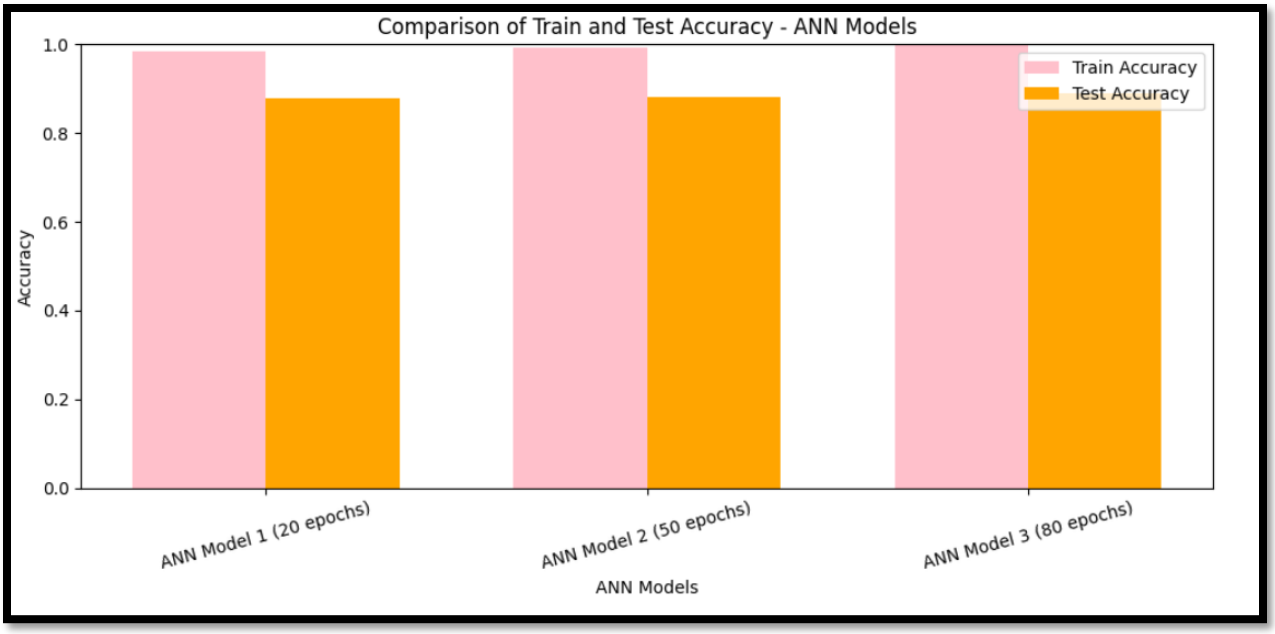
Model 1



Model 2



Model 3



PERFORMANCE OF IMPLEMENTED ALGORITHMS:

K-nearest Neighbor (KNN):

The K-Nearest Neighbors (KNN) model exhibits overfitting around k=2, most likely due to the model's excessive sensitivity to the training data and its small number of neighbors, which leaves it extremely vulnerable to noise and slight variations in the data. High variation and inadequate generalization to the test set are the outcomes of this. Nevertheless, the model performs noticeably better when k rises to 5 and 15, with training and test accuracy being more evenly distributed. The model may take into account a wider range of neighbors when k is larger, which smooths out the predictions and enhances its generalization capabilities. The model gets more stable and less susceptible to outliers as k rises, which improves its generalization and lessens overfitting

Logistic Regression:

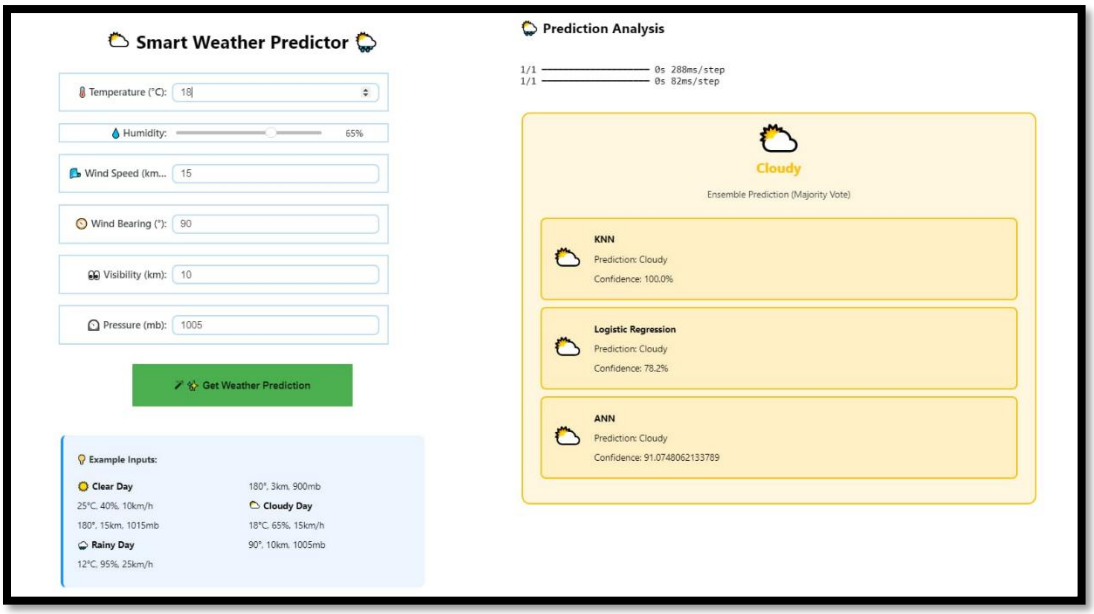
The Logistic Regression models across all three configurations are yielding accuracies in the 60s, which suggests that the models are experiencing underfitting. Underfitting occurs when the model is too simple to capture the underlying patterns in the data, resulting in poor performance both on the training and test sets. In this case, the Logistic Regression models may not be complex enough to fully represent the relationship between the input features and the target variable, leading to low accuracy.

Artificial Neural Network (ANN):

All three ANN models show evidence of overfitting, according to the analysis of the training and validation accuracy and loss graphs. While the training accuracy and training loss both continuously rise, the validation accuracy fluctuates and does not increase steadily, and the validation loss initially drops but then fluctuates in subsequent epochs. These behaviors show that the models are having trouble generalizing to unknown validation data, despite learning the training data too well. The variations in validation metrics imply that the models are learning particular patterns in the training data rather than generalized patterns, even though training accuracy is improving with each model.

USER INTERFACE:

This interface is a Smart Weather Predictor that allows users to input various weather-related parameters, such as temperature, humidity, wind speed, and pressure. Based on these inputs, the model makes weather predictions using multiple algorithms, including KNN, Logistic Regression, and ANN. The final prediction is determined by an ensemble method (majority vote), with the interface displaying the most likely weather condition and the confidence level of each model.



SPLITING THE TEST SET BY 30%:

By changing the test spilt from 20% to 30% there is no difference in the performance of K-Nearest Neighbor (KNN) and Logistic Regression but a slight change can be notice in train and test accuracy of Artificial Neural Network (ANN), it is giving slightly better accuracies.

K-Nearest Neighbor (KNN):

Model	Hyperparameters	Train Accuracy	Test Accuracy
1	k = 2	87.96%	78.60%
2	k = 5	85.76%	78.89%
3	k = 15	79.33%	77.02%

Logistic Regression:

Model	Hyperparameters	Train Accuracy	Test Accuracy
1	Multi-class = 'multinomial', solver = 'lbfgs', max_iter = 1000	61.20%	61.81%
2	Multi-class = 'ovr', solver = 'saga', penalty = 'l1', C = 0.5, max_iter = 2000	60.81%	61.03%
3	Multi-class = 'multinomial', solver = 'newton-cg', penalty = 'l2', C = 1.5, max_iter = 3000	61.21%	61.80%

Artificial Neural Network (ANN):

Model	Hyperparameters	Train Accuracy	Test Accuracy
1	Epochs = 20 & batch_size = 100	75.72%	74.61%
2	Epochs = 50 & batch_size = 130	74.44%	73.63%
3	Epochs = 80 & batch_size = 150	78.40%	76.84%